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As $k \to \infty$; $(0.92)^k \to 0$

$X_k = c_1 v_1$

$X_k = 0.125 \begin{bmatrix} 3 \\ 5 \end{bmatrix}$

$c_1 = 0.125$

So $\Rightarrow \begin{bmatrix} 0.375 \\ 0.625 \end{bmatrix}$ Ans.

To find c_1 and c_2

~~Mod~~

Take an iverse of P_B (P_B^{-1}) and multiply by X_0 given vector.

Dot Product.

$u, v \in \mathbb{R}^n$

$u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$; $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$

$\because u^T = [u_1 \; u_2 \; u_3 \; \dots \; u_n]$

$u^T v = [u_1 \; u_2 \; u_3 \dots u_n] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ \vdots \end{bmatrix}$

$= u_1 v_1 + u_2 v_2 + u_3 v_3 \dots$

$= u \cdot v$

It is also called inner product and the result is a scalar.

$= u^T \cdot u = u \cdot u = u_1^2 + u_2^2 + u_3^2 + u_n^2$

$u \cdot u \geq 0$ (zero iff $u = 0$)

$\Rightarrow$ Dot product is commutative

$u^T v = v^T u$

$\Rightarrow$ Dot product is distributive.

$(u+v) \cdot w = u w + v \cdot w$

$\Rightarrow c u \cdot v = c(u \cdot v)$

$\Rightarrow$ length of 'u'

it is the sum of the squared values of a vector and its sqrt.

$\|u\| = \sqrt{u_1^2 + u_2^2 + u_3^2 + \dots u_n^2} = \sqrt{u \cdot u}$

$\|u\| = \|u\|_2$ {some notation}

"norm" $\{\|u\|\}$.

= subscript '2' is called/used for Euclidean norm.

$\Rightarrow$ Square of a norm is

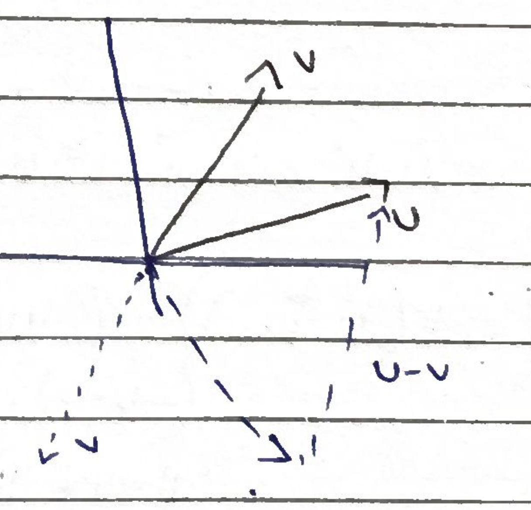
$\|u\|^2 = u \cdot u$

$\Rightarrow$ Distance b/w two vectors

dist (u, v).

$\Rightarrow$ Difference of 2 vectors and then take a norm of resultant vector.

so, $\|u - v\|$



$\Rightarrow u + (-v)$

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$\sqrt{(u-v)\cdot(u-v)}$

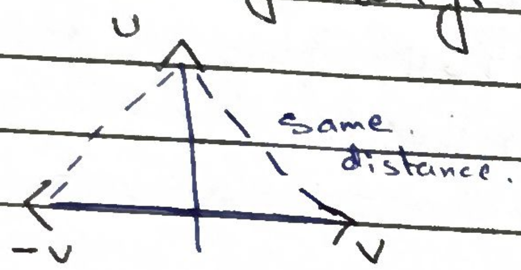
we consider $u-v$ as single vector and we know the norm is the dot product of its own self and then sqrt of it.

so,

$= \sqrt{(u-v)\cdot(u-v)}$

$= \sqrt{(u_1-v_1)^2+(u_2-v_2)^2+(u_3-v_3)^2 \cdots +(u_n-v_n)^2}$

Orthognality.



If the $-v$ is also $\perp$ to the 'u' then the distances between them will besame

$= (\text{dist}(u,v))^2 = (\|u-v\|)^2 = (u-v)\cdot(u-v)$

square of norm = dot product

$= u\cdot u - u\cdot v - u\cdot v + v\cdot v$

$= \|u\|^2 + \|v\|^2 - 2u\cdot v$

$=$

$(\text{dist}(u, -v))^2 = (\|u+v\|)^2$

$= (u+v)\cdot(u+v)$

$= \|u\|^2 + \|v\|^2 + 2u\cdot v$

$((\text{dist}(u,v))^2 = (\text{dist}(u,-v))^2$

$\cancel{\|u\|^2+\|v\|^2} - 2u\cdot v = \cancel{\|u\|^2+\|v\|^2} + 2u\cdot v$

$-2u\cdot v = 2u\cdot v$

$-2(u\cdot v) = 2(u\cdot v)$

$\Rightarrow u\cdot v = 0$

It is only applicable means proving the $\perp$ when $u\cdot v = 0$

$\Rightarrow \|u+v\|^2 = \|u\|^2 + \|v\|^2$

pythagores theorem

$\Rightarrow$ Establishing orthognality in multiple vectors.

Orthognal Complements

let say 'w' is a subspace of $\mathbb{R}^n$ {means we have taken a subset of vectors from $\mathbb{R}^n$}

$\Rightarrow W^\perp$ {w is orthognal component}

if any vector 'u' in $W \Rightarrow u\cdot v = 0$ $v \in W^\perp$

{vectors from two subspaces are $\perp$ to each other and are orthognal components}

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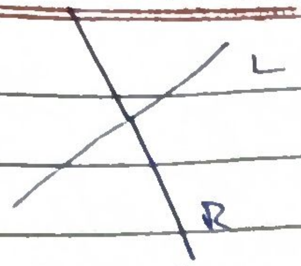

=> The R is $\perp$ to L so its orthognal

=> Zero vector is $\perp$ to every vector and it's self. and it is also an orthognal component.

=> Matrix $A_{m\times n}$ its null space is of $\mathbb{R}^n$.

=> $(\text{Null } A)^{\perp} \in \mathbb{R}^n$

$A.x = 0$.

=> Rowspace A is orthognal complements

=> $((\text{Null } A)^T)^{\perp} \longleftrightarrow \text{Col } A$.

$A^T x = 0$.

Orthognal Sets

$S = \{u_1, u_2, u_3 \cdots u_p\}$.

This set of vector is orthognal set only when $u_i \cdot u_j = 0$; for $i \neq j$.

means for every pair of vector.

$u_1 \cdot u_2 = 0$ then it is orthognal set. $u_i \in \mathbb{R}^n$

=> Basis is a machine which generates vectors.

=>

$y \in \mathbb{R}^n$

$y = c_1u_1 + c_2u_2 + c_3u_3 + \cdots + c_pu_p$

Form a matrix equation of it.

$$U = \begin{bmatrix} | & | & | & | & & | \\ u_1 & u_2 & u_3 & u & \cdots & u_p \\ | & | & | & | & & | \end{bmatrix}$$

$$c = \begin{bmatrix} c_1 \\ c_2 \\ c \\ \vdots \\ c_p \end{bmatrix}$$

then

$Uc = y$.

=> we want to find beta coordinates of y.

$y = [U]_B . (y)_B$.

$[y]_B = [U_B]^{-1} y$.

=> We can find the value of constant without row reducing and other by this way.

$$c_1 = \frac{y \cdot u_1}{u_1 \cdot u_1}$$

$\vdots$

$$c_p = \frac{y \cdot u_p}{u_p \cdot u_p}$$

There will be unique values and no free variables. But it is the system is consistent cuz basis are consisted.

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cuz we are dealing with basis.

$\Rightarrow S = \{e_1, e_2, e_3\}\ \mathbb{R}^2$

express 'y' as linear combination of e_i's.

$e_1 = \begin{bmatrix}1\\0\\0\end{bmatrix};\ e_2 = \begin{bmatrix}0\\1\\0\end{bmatrix},\ e_3 = \begin{bmatrix}0\\0\\1\end{bmatrix}$

$y = \begin{bmatrix}1\\2\\3\end{bmatrix}$

$y = e_1c_1 + e_2c_2 + e_3c_3$

$c_1 = \dfrac{y \cdot e_1}{e_1 \cdot e_1} = \dfrac{y \cdot e_1}{1} = \begin{bmatrix}1\\2\\3\end{bmatrix}\begin{bmatrix}1\\0\\0\end{bmatrix} = 1c_1$

$c_2 = \dfrac{y \cdot e_2}{e_2 \cdot e_2} = \dfrac{y \cdot e_2}{1} = \begin{bmatrix}1\\2\\3\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} = 2 = c_2$

$c_3 = \dfrac{y \cdot e_3}{e_3 \cdot e_3} = \dfrac{y \cdot e_3}{1} = \begin{bmatrix}1\\2\\3\end{bmatrix}\begin{bmatrix}0\\0\\1\end{bmatrix} = 3 = c_3$.

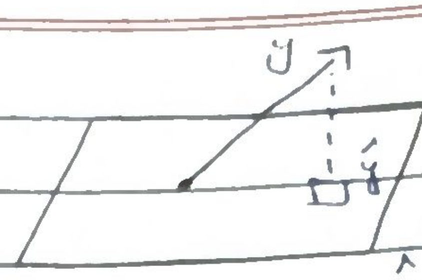
e_1, e_2, e_3 are unit vector so dot product of them = '1'

$\boxed{\text{unit vector} = \dfrac{u}{\|u\|}}$

so, $c_1 = 1, c_2 = 2, c_3 = 3$

$y = 1\begin{bmatrix}1\\0\\0\end{bmatrix} + 2\begin{bmatrix}0\\1\\0\end{bmatrix} + 3\begin{bmatrix}0\\0\\1\end{bmatrix}$

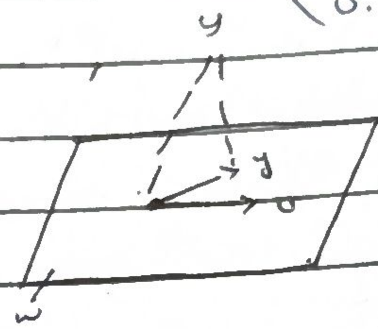
$\begin{bmatrix}1\\2\\3\end{bmatrix} = \begin{bmatrix}1\\2\\3\end{bmatrix}$.



$\text{Proj}_u\, y = \hat{y}$

$= \alpha u$

$= \left(\dfrac{y \cdot u}{u \cdot u}\right) u$



$\hat{y} = \left(\dfrac{y \cdot u_2}{u_2 \cdot u_2}\right)u_2 + \left(\dfrac{y \cdot u_1}{u_1 \cdot u_1}\right)u_1$

$u_i \cdot u_j = 0 \quad \because i \neq j$

Rescale

$\dfrac{u_i}{\|u_i\|}$

$V = \begin{bmatrix} | & & | \\ v_1 & \cdots & v_p \\ | & & | \end{bmatrix}$

$v_i = \dfrac{u_i}{\|u_i\|}$

$\text{Proj}_W\, y = \left(\dfrac{y \cdot v_1}{v_1 \cdot v_1}\right)v_1 + \dfrac{y \cdot v_2}{(v_2 \cdot v_2)} + \dots \dfrac{y \cdot v_p}{v_p \cdot v_p}$

$\hat{y}$ projection vector.

$\Rightarrow (y \cdot v_1)v_1 + (y \cdot v_2)v_2 + \dots y \cdot v_p(v_p)$

$v_1 \cdot v_1 = v_2 \cdot v_2 = 1$ (unit matrix)

$= V V^T y$.

V = matrix $= \begin{bmatrix} | & | & | & & | \\ v_1 & v_2 & v_3 & \cdots & v_p \\ | & | & | & & | \end{bmatrix}$

V^T = scalar set $= \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$

$= \begin{bmatrix} | & | & | & & | \\ v_1 & v_2 & v_3 & \cdots & v_p \\ | & | & | & & | \end{bmatrix}\begin{bmatrix} v_1^T \cdot y \\ v_2^T \cdot y \\ \vdots \\ v_p^T \cdot y \end{bmatrix}$

eigen vectors just scale.

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$= V^T V$

$V_{n \times p} : V^T_{p \times n}$

$V^T . V = V_{p \times p}$

→ It result in a scalar.

if v is an orthonormal matrix $n \times p$. then.

$V^T . V = I_p$.

Properties of orthonormal matrix.

i) $\| Vx \| = \| x \|$

norm is preserved of it.

before and after length will be same.

(ii) $(V.x).(Vy) = x.y$.

(iii) $(V.x).(V.y) = 0$

iff x and y $\perp$ to each other.

GRAM-SCHMIDT Process

$S = \{ x_1, x_2, \ldots x_p \}$ for $W \in \mathbb{R}^n$

project v_1 on v_2 and then project v_3 on the projected v_1 and v_2 and subtract it from them it would be $\perp$

$V_1 = x_1$

$V_2 = x_2 - \left(\frac{x_2 . v_1}{v_1 . v_1} \right) v_1$

Now $v_1 \perp v_2$

$V_3 = x_3 - \left(\frac{x_3 . v_1}{v_1 . v_2} \right) v_1 - \left(\frac{x_2 . v_1}{v_1 . v_1} \right) v_2$

$\vdots$

$V_p = x_p - \left(\frac{x_p . v_1}{v_1 . v_1} \right) v_1 - \cdots - \left(\frac{x_p . v_{p-1}}{x_{p-1} x_{p-1}} \right) . v_{p-1}$

$\text{span}\{ v_1, v_2, \ldots v_p \} = \text{span}\{ x_1 \ldots x_p \}$.

if we want to get orthonormal basis divide each one with unit vector. / by their norm.

Convert into orthonormal.

$A_{m \times m} = Q_{m \times n} . R_{n \times n}$

$Q^T A = R_{n \times n}$

↳ upper Δ matrix

it is called QR factorization.

Question.

$v_1 = \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix}; \; v_2 = \begin{bmatrix} -1 \\ 1 \\ -2 \end{bmatrix};$

$y_3 = \begin{bmatrix} -9 \\ 1 \\ 6 \end{bmatrix}$ $w = \text{span}\{ v_1, v_2 \}$

$\text{Proj}_w \, y = ?$

Sol.

$\hat{y} = \left(\frac{-9 . (-7)}{-7 . -7} \right) \begin{bmatrix} -7 \\ 1 \\ 4 \end{bmatrix} +$

$= A (A^T A)^{-1} A^T y$.

$A = \begin{bmatrix} -7 & -1 \\ 1 & 1 \\ 4 & -2 \end{bmatrix}$

Check for orthognality.

$= (-7)(-1) + (1)(1) + (4)(-2)$

$= \{ \text{orthogonal} \}$

$= 7 + 1 - 8 = 0$ Ans

$A^T \cdot A = \begin{bmatrix} -7 & 1 & 4 \\ -1 & 1 & 2 \end{bmatrix}\begin{bmatrix} -7 & -1 \\ 1 & 1 \\ 4 & 2 \end{bmatrix}$

$= \begin{bmatrix} 66 & 16 \\ 16 & 6 \end{bmatrix}$

Now,

$(A^T \cdot A)^{-1} = \frac{1}{140}\begin{bmatrix} 6 & -16 \\ -16 & 66 \end{bmatrix} = \frac{1}{70}\begin{bmatrix} 3 & -8 \\ -8 & 33 \end{bmatrix}$

$\det(A^T A) = 66.6 - (16)(16)$

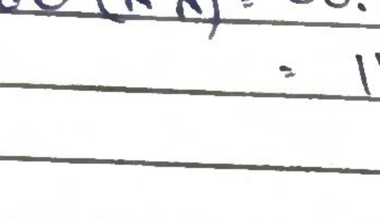
$= 140.$

Distance $= \sqrt{(y-\hat{y}).(y-\hat{y})}$

$= \| y - \hat{y} \|.$

Best Approximation

•) $y \notin W$ (not a part).



many vectors can be best approximation, but $\hat{y}$ is the most best.

$\|y - \hat{y}\| < \|y - v\|$

This norm is near.

$\|y - v\|^2 = \|\hat{y} - v\|^2 + \|y - \hat{y}\|^2$

By pythogores theorem.

The best approximation is the nearest distance and the one which has less norm.

$\|\hat{y} - v\| \neq 0.$

$\hat{y} - v$ is not the best approximation.

$\Rightarrow$ least squared Solution.

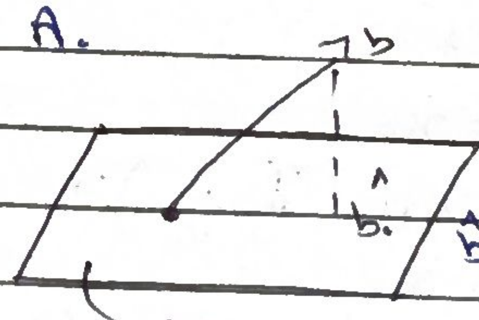
$Ax = b.$

if we have b and x we can find A.

and when it is not applied {pivot in agumented col}

$\Rightarrow b \notin \text{col } A \Rightarrow Ax \neq b.$

cuz there is no 'x' which with A gives b so it is not in column space of A.



$\hat{b} = A\hat{x}$

least squard solution $\hat{x}$ for $Ax = b$ is $A\hat{x} = \hat{b}$

such that $\|b - A\hat{x}\| \leq \|b - Ax\|$

it minimizes this norm.

$\Rightarrow$ if $b \in \text{col } A$ so it is equal

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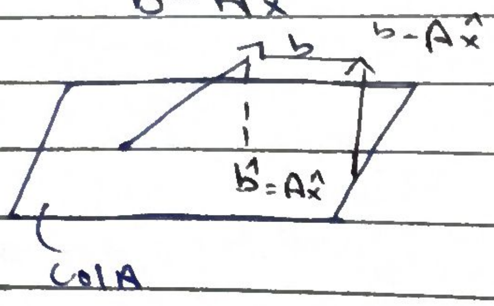
solution will exist everytime.

the consistent system.

$A\hat{x} = \hat{b}$

$b - A\hat{x}$



So,

- $a_j \cdot (b - A\hat{x}) = 0.$
- $A^T (b - A\hat{x}) = 0.$
- $A^T b - A^T A \hat{x} = 0.$

$\boxed{A^T b = A^T A \hat{x}}$ - normal equation.

- Is it a normal equation.
- It is a consitent system and have a solution which concides with least squared solution.